

OZONE**ICSC: 0068 (April 2009)****CAS #: 10028-15-6****EC Number: 233-069-2**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible but enhances combustion of other substances. Many reactions may cause fire or explosion. Risk of fire and explosion on contact with combustible substances.	NO open flames, NO sparks and NO smoking. NO contact with combustible substances. Closed system, ventilation, explosion-proof electrical equipment and lighting.	In case of fire in the surroundings, use appropriate extinguishing media. Combat fire from a sheltered position.

STRICT HYGIENE!			
	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Cough. Headache. Shortness of breath. Laboured breathing.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Refer for medical attention.
Skin	ON CONTACT WITH LIQUID: FROSTBITE.	Cold-insulating gloves.	ON FROSTBITE: rinse with plenty of water, do NOT remove clothes. Refer for medical attention .
Eyes	Redness. Pain.	Wear face shield or eye protection in combination with breathing protection.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion			

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Consult an expert! Personal protection: chemical protection suit including self-contained breathing apparatus. Ventilation.	According to UN GHS Criteria  DANGER May cause or intensify fire; oxidizer Fatal if inhaled Causes eye irritation Causes damage to lungs if inhaled Causes damage to the lungs through prolonged or repeated exposure if inhaled Transportation UN Classification
STORAGE	
Fireproof if in building. Separated from all substances. Cool.	
PACKAGING	



International
Labour
Organization



World Health
Organization

Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.
© ILO and WHO 2021



European
Commission

OZONE**ICSC: 0068****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS OR BLuish GAS WITH CHARACTERISTIC ODOUR.

Physical dangers

The gas is heavier than air.

Chemical dangers

Decomposes on warming. This produces oxygen. This generates fire and explosion hazard. Reacts violently with inorganic and organic compounds. This generates fire and explosion hazard. Attacks rubber.

Formula: O₃

Molecular mass: 48.0

Boiling point: -112°C

Melting point: -193°C

Solubility in water: none

Relative vapour density (air = 1): 1.6

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation.

Effects of short-term exposure

The substance is irritating to the eyes and respiratory tract. The substance may cause effects on the central nervous system. This may result in impaired vigilance and performance. Inhalation of the gas may cause lung oedema. See Notes. The effects may be delayed. The liquid may cause frostbite.

Inhalation risk

A harmful concentration of this gas in the air will be reached very quickly on loss of containment.

Effects of long-term or repeated exposure

Repeated or prolonged inhalation of the gas may cause effects on the lungs.

OCCUPATIONAL EXPOSURE LIMITS

TLV: (light work): 0.1 ppm as TWA; (moderate work): 0.08 ppm as TWA; (heavy work): 0.05 ppm as TWA.

TLV: (less than 2 hours work): 0.2 ppm as TWA.

TLV: A4 (not classifiable as a human carcinogen).

MAK: carcinogen category: 3

ENVIRONMENT

This substance may be hazardous to the environment. Special attention should be given to plants.

NOTES

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.

Immediate administration of an appropriate inhalation therapy by a doctor, or by an authorized person, should be considered.

ADDITIONAL INFORMATION**EC Classification**

All rights reserved. The published material is being distributed without warranty of any kind, either expressed or implied. Neither ILO nor WHO nor the European Commission shall be responsible for the interpretation and use of the information contained in this material.